

Session 1: Industry Analysis (and Strategy ID)

Competitive Strategy

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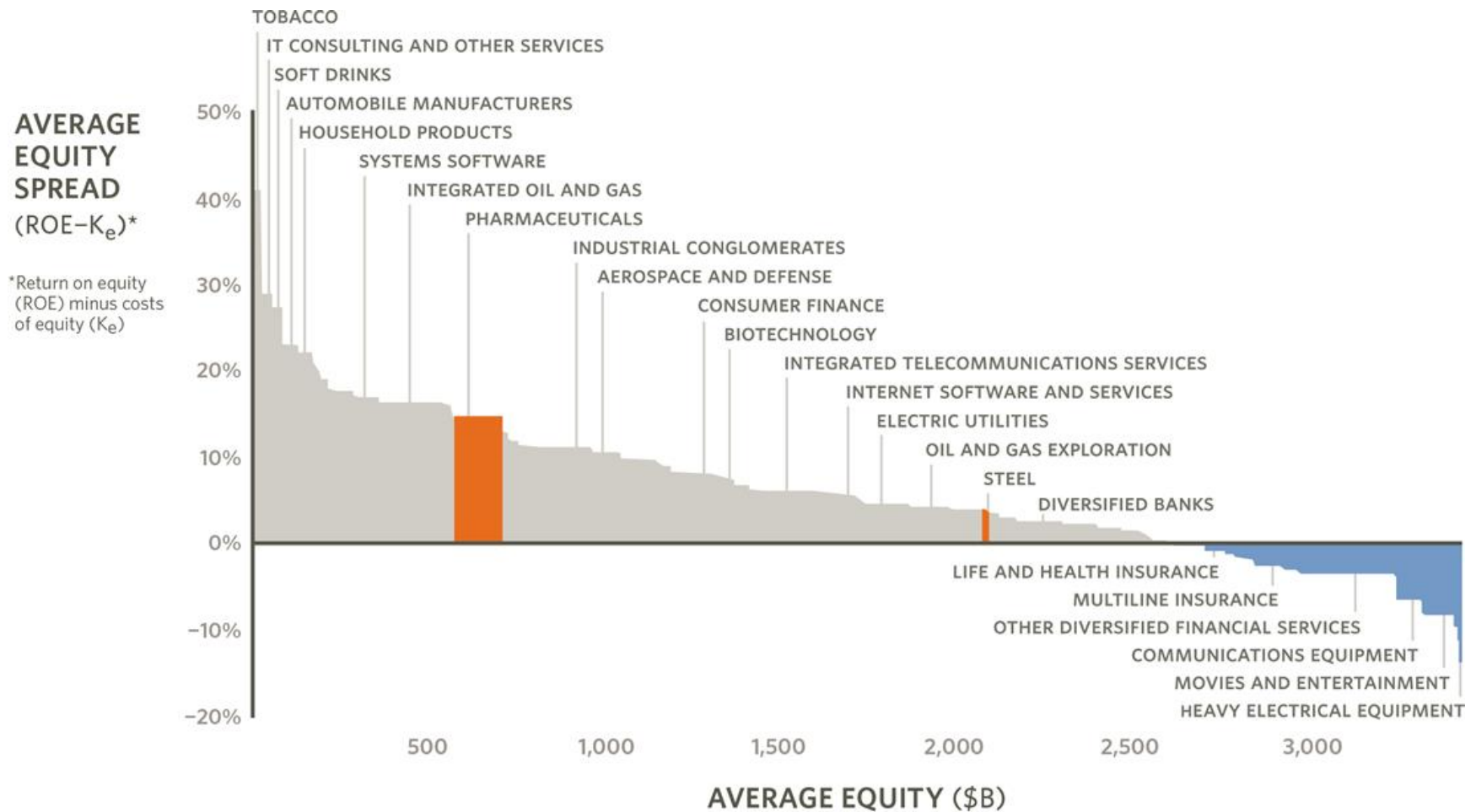


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Across-industry variation

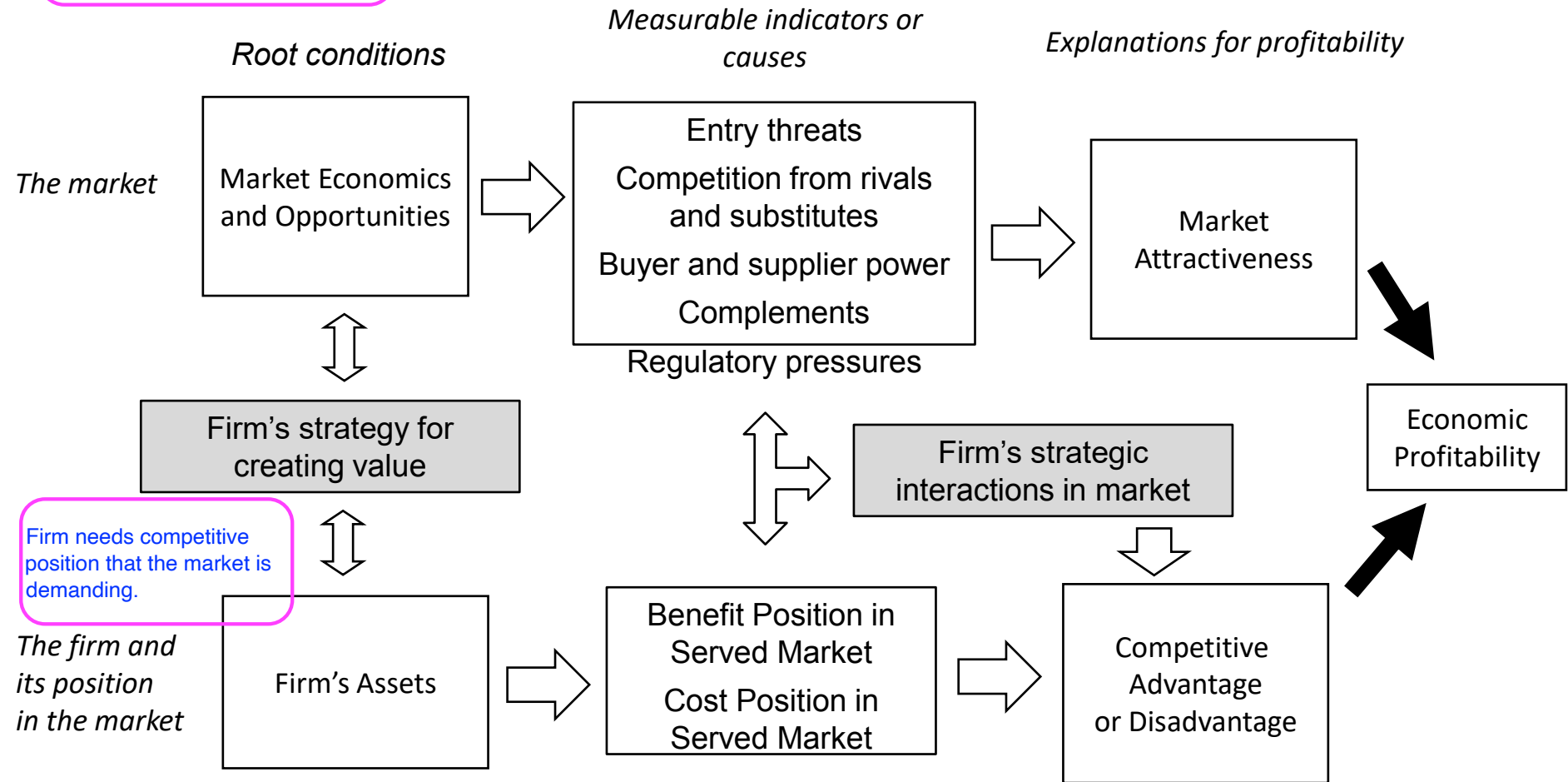
- Motivating stylized fact: there are large, persistent differences in profitability across industries
- Illustration
 - Plot measure of “economic profit” by industry
 - Assume *economic profit* $\propto$ *return on equity* – *cost of capital*
 - Restrict attention from 1994 to 2012

Result [Ghemawat and Rivkin, 2014]



A Framework for Competitive Strategy Analysis

Main framework for today and tomorrow (21-22 Jan)



What determines industry profitability?

Any framework must answer three questions

1) Is the product valuable?

Value Creation

2) Is the value being dissipated by competition?

Value Capture: Actual and potential rivalry

how do we avoid rivals / new entrants from capturing this value created?

3) Is the value being captured somewhere else in the supply chain?

Value Capture: supplier and buyer power

How to retain the value created instead of slipping to up/downstream. Many times, industry create value, but it is captured by someone else.

example; is there supplier power? Ricardian Rent concept.

e.g, TSMC, ASML.

A Framework for Industry Analysis

Value Creation: what is PIE

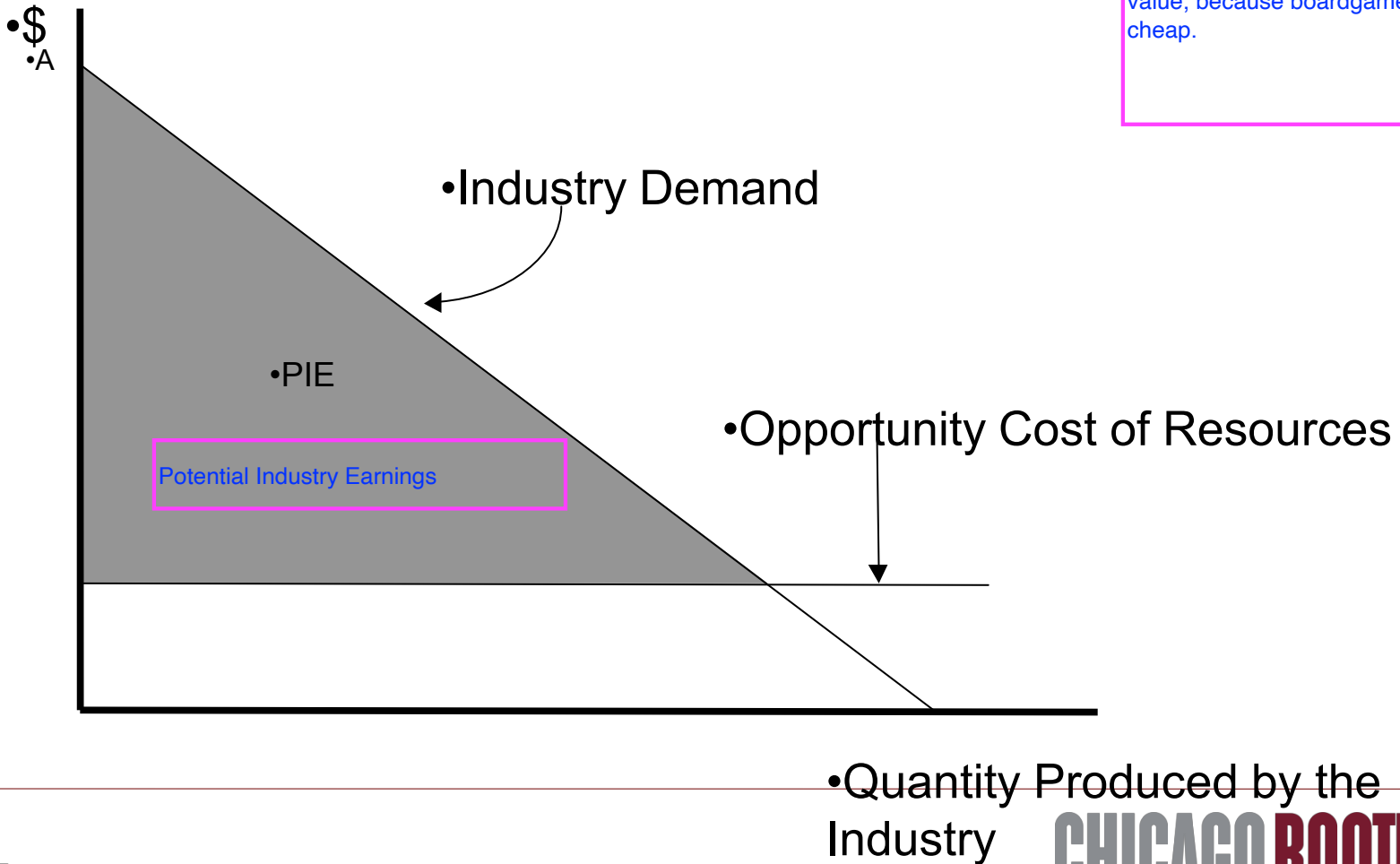
Potential Industry Earnings

Value Capture:

- Actual and potential rivals (rivalry, entry)
- Other players in supply chain (suppliers and buyers)

Identifying PIE

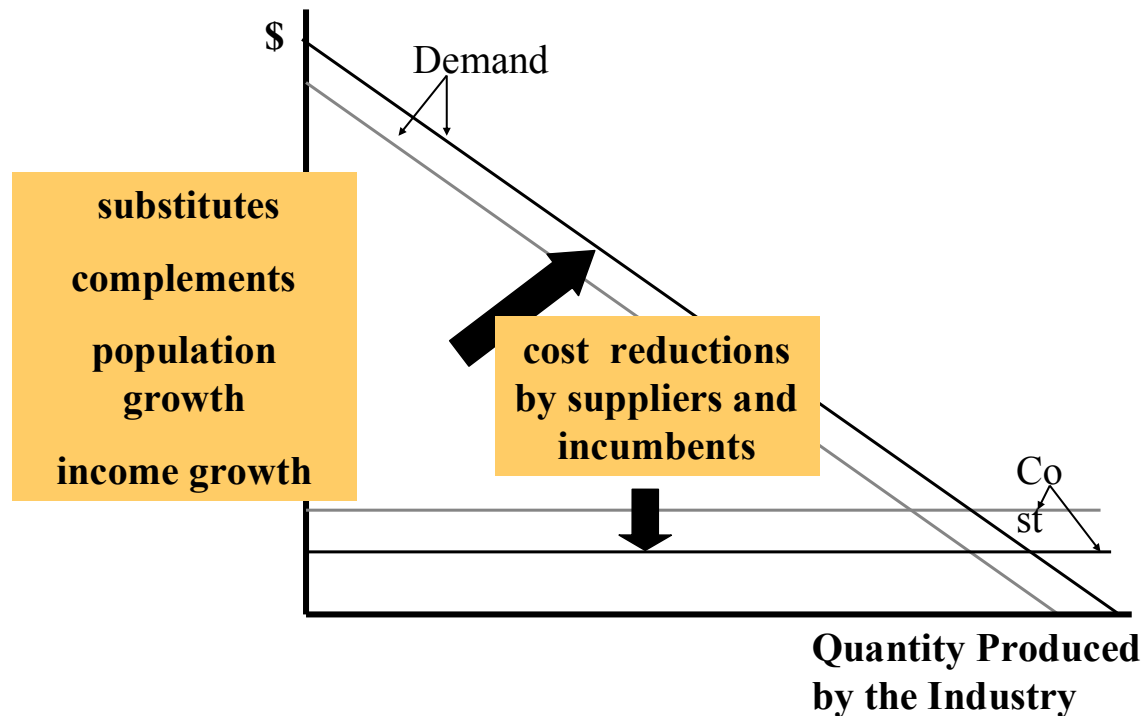
2007; he worked with Philips. They are working on a touchscreen device that allows users to play board games. Machine cost \$5000.
- need to check consumer WTP.
- does this machine creates value?
Analysis. This machine creates totally no value, because boardgames are very cheap.



Factors that influence value creation

Demand; total addressable market size.
Accessible? obtainable? is my value
created easily substituted?
- note; substitutes may severely limit your
ability to make profits.

Can technology reduce my cost?



Substitutes

example;
- Airline vs Zoom (substitute)
- Other electronic devices vs Smart Phones

Key Question

To what extent does competition from substitute products outside the defined boundaries of the industry erode the profitability of a typical firm in the industry?

Substitutes limit the profits of an industry by placing a ceiling on industry prices.

- The more attractive the price-performance tradeoff offered by substitutes, the more important the substitute product.

Complements

Two business X and Y are complements if the consumer's willingness to pay (or valuation) for the product sold by X increases when the cost to the consumer of Y decreases.

- Examples:
 - iTunes iPhone
 - Golf courses and real estate
 - Windows Intel

A Framework for Industry Analysis

Value Creation: what is PIE

Potential Industry Earnings

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4 aspects that affect our ability to capture?

- rivalry (competitors)
- new entrants
- people who buy from us
- people who sell to us

I. Rivalry (1)

Bertrand Model:

One shot competition, homogeneous goods and constant unit costs.

Results in prices equal to unit costs

3 main different strategies

How do we get away from this nasty world?:

1. Capacity constraints
2. Differentiated products/switching costs
3. Repeated interaction – tacit coordination

Rivalry (2): Role of Capacity

Excess capacity is a key driver of price cuts

- Provides ability and motivation to reduce prices

Need for utilization of existing capital investments

- Particularly relevant the more the industry is one with high fixed costs/low marginal costs

Rivalry (3): Differentiation



•Dark
beer



•Light
beer

Rivalry (3): Differentiation

differentiation will soften price competition but will increase fixed costs.

question; is there a concept as to too much differentiation?

Differentiation reduces price competition

- Decreasing responsiveness of demand to prices decreases incentive to lower prices

Switching costs/search costs also decrease responsiveness to price

- Thus they also reduce price competition

May create differentiation/switching costs on purpose

- Will see several examples during course
- Brand loyalty plays similar role

•Dark
beer

•Light
beer

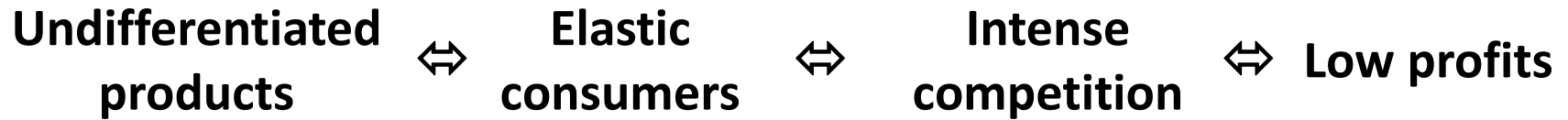
Differentiation: How to measure?

- Consider two otherwise identical markets, each with two firms
- In each, firms start at an arbitrarily high price (e.g., monopoly p^*)
- In each, one firm lowers price a certain amount
- *In Market A, they consider considerable market share*
- *In Market B, they gain very little*

Differentiation: How to measure?

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- In each, one firm lowers price a certain amount
- In Market A, they gain considerable market share
- In Market B, they gain very little
 - *In which market are prices and profits higher?*
 - *In which market are the products more “differentiated”?*
 - *In which market is competition more “intense”?*
 - *What parameter governs the market share changes?*

Differentiation: How to measure?



Example: some consumer products

- *Consider a 10% price increase*
- Ask, “how much market share is lost”(i.e., a lot or a little)?
- Ask, “to whom is it lost?”
- Examples
 - *Apple iPhone*
 - *Samsung television*
 - *Plavix (medicine to prevent blood clots after a heart attack, on patent)*
 - *Bumble Bee tuna*
- *In which do we see high and low profits?*

Solution to earlier problems

- We now have a *measure* of “rivalry”:
(i.e., cross-product price elasticity, a.k.a. “cross-price elasticity”)
- We also have a *measure* of “substitution”
 - Implementation
 - We can write down all the alternatives from which we steal market share when we lower price
 - Some will stick out as “outside alternatives”
 - Example
 - Luxury cars compete with other cars (mostly luxury ones)
 - Compact cars compete with other compacts but also with public transport, etcetera!

Rivalry (3): Tacit Coordination (much more later)

Conditions that facilitate tacit coordination

Concentrated market

Facilitating practices used

Public sales terms; firms have good info. about competitors

Small, frequent orders

Strong capacity utilization

Rapidly growing demand

Exit barriers increase rivalry

unprofitable firms stick around

II. Barriers to Entry

The key question

To what extent does the threat or incidence of entry erode the profitability of a typical firm in the industry?

Factors that restrict entry into an industry:

- Economies of Scale
- Large Sunk Expenditures
- Learning by doing
- Branding
- Patents and licenses
- Expectation of aggressive response to entry (later)

When does 'capital' constitute a BTE?

When it results in large economies of scale

- But large capital expenditures are neither necessary nor sufficient for there to large economies of scale
- Key: Minimum Efficient Scale (MES) relative to market size

When investment is large and 'sunk'

Economies of Scale and MES

Production economies

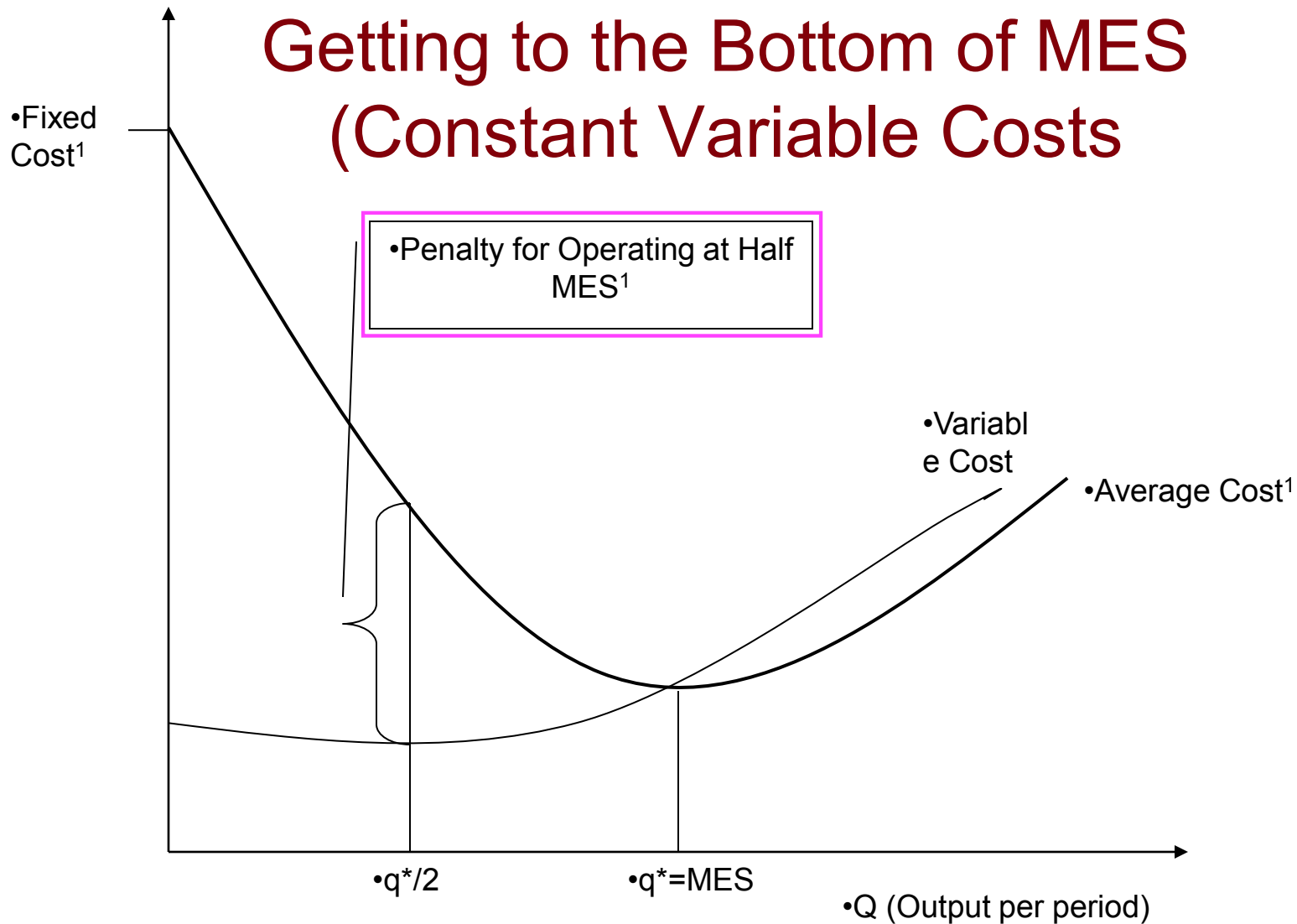
- Due to lumpy capital, specialization of labor/ capital, technological factors.
- Best measured by comparing minimum efficient scale to market size (approximates number of efficient scaled firms that can be sustained)
 - How many firms “fit”? $N = \text{Market size} / \text{MES}$ (e.g. CCS)

Walmart, town is big enough for 1 store, but not 3.
- any new entrant will spoil the market.

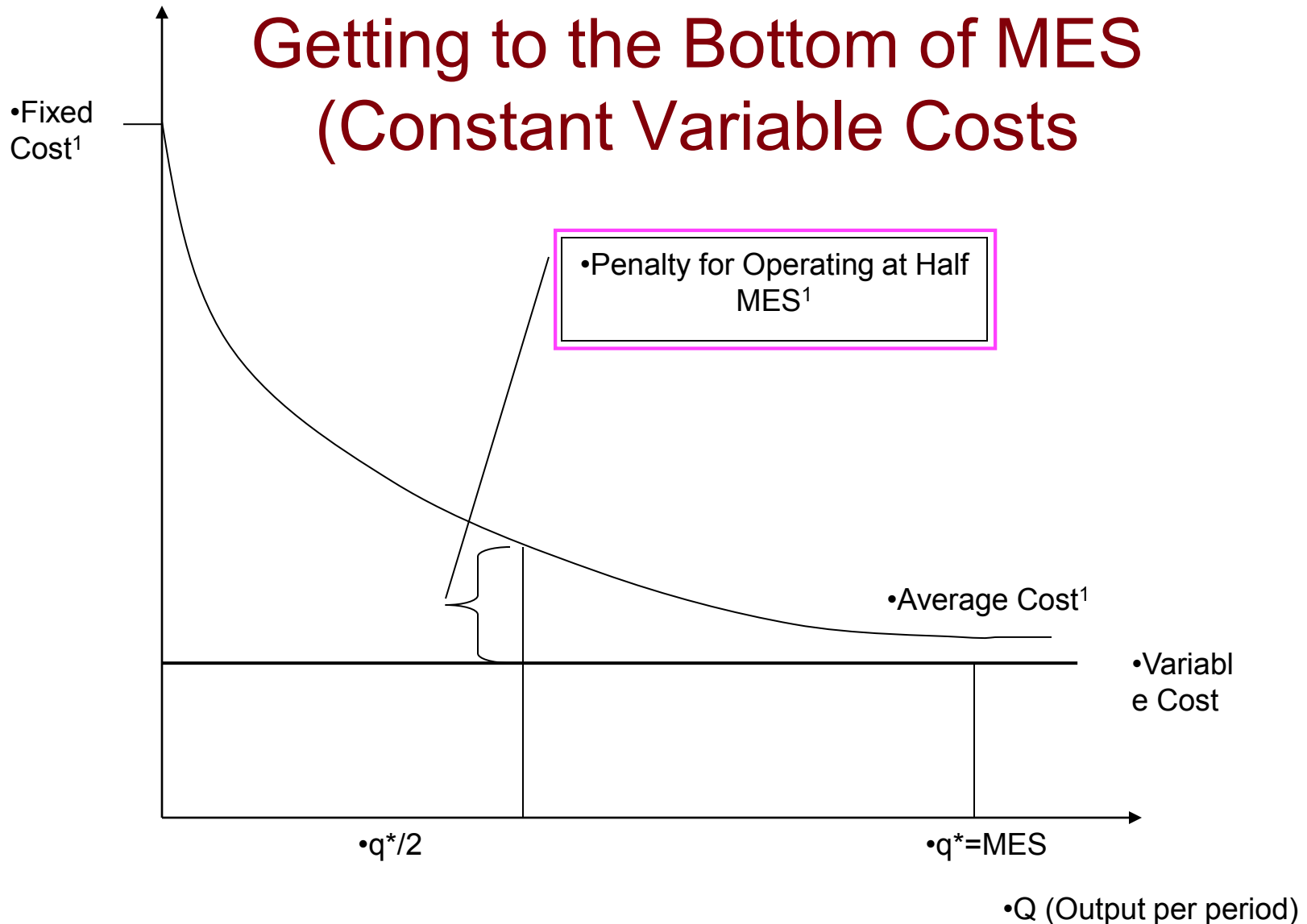
“Lumpiness” means incumbents can make profits but any additional entry would “spoil the market”.

Getting to the Bottom of MES

(Constant Variable Costs)



Getting to the Bottom of MES (Constant Variable Costs)



Sunk Costs

example; animated movies. usually only 1-2 succeed a year. high fixed costs, competitors are deterred.

If investment made is specific to opportunity and firm cannot recover it.

- To the extent that investment made is irreversible, hit and run entry impossible

Large sunk investment is a barrier to entry, since if exit is costly, more unlikely to enter under uncertainty

Some firm investments create barriers to entry by increasing the amount of sunk investments required to enter industry ((much more later)

- Investments in brand building (advertising) -- entirely sunk
- Investments in R&D

Example: commercial aircraft manuf.

- Only 2 wide-body jet manufacturers
- ~1 new design every 15-20 years
- Examples
 - Boeing 787: launched 2007 at \$32 billion design cost
 - Airbus A350: launched 2010 at \$15 billion design cost
- Economies of scale are not a mere theoretical constructs:
A350 cost several times the *combined* profits of the next two biggest *firms* (Embraer and Bombardier) over the last 5 years
- **Barrier to entry: large fixed and/or sunk costs**

II. Learning-by-doing

- Refers to how costs fall with accumulated production
- Usually thought of as manufacturing relationship
- Much broader and more nuanced (e.g., online search...)

Example: search engines

- More market share = more searches = better results in future
- Bing launched in 2009
 - Loses \$1 billion per quarter circa 2011(!)
 - Gains market share so that by October 2015
 - Google = 64%
 - Bing = 34%
- **Barrier to entry: LBD (i.e., accumulated “know-how”)**
- *Note that LBD is dynamic, so it’s distinct from scale economies*

III. Branding

- Refers to how the value of products rise with accumulated advertising/marketing effort
- Less mechanical and more psychological than LBD
- Yet, no less important

Example: luxury sedans



Mercedes S550



Hyundai (Genesis) G80

Price? Sales??

Example: luxury sedans

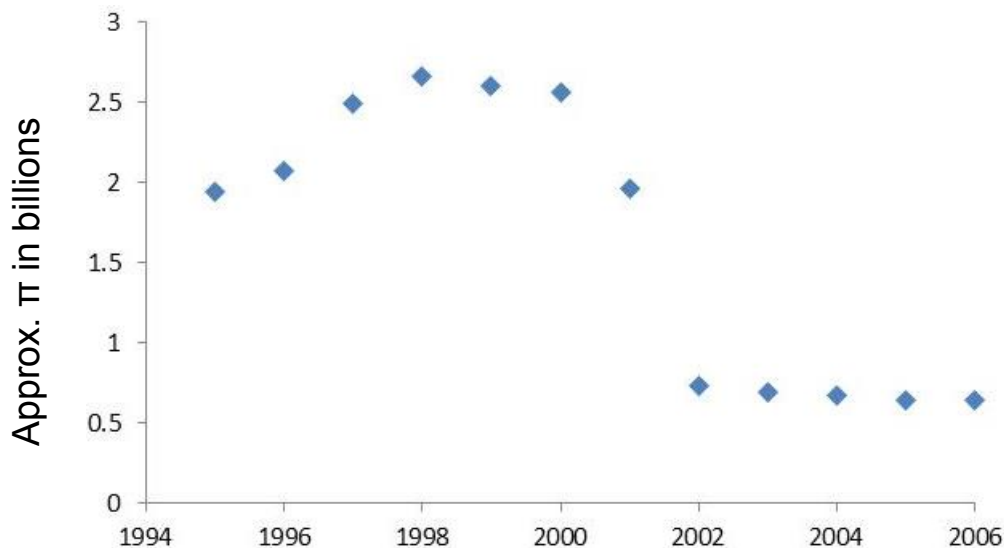


How large is $|dQ_{\text{Mercedes}} / dP_{\text{Genesis}}|$?
(How elastic are Mercedes consumers?)

IV. Exclusive rights to make/use/sell an idea

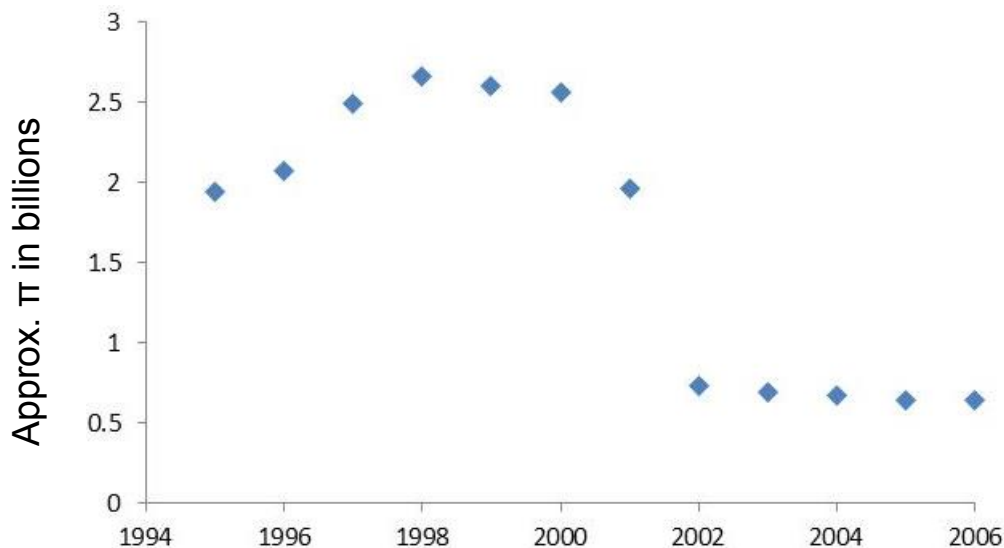
- Intellectual property conveys certain rights to individuals
- Types of “IP”
 - **Patents** exclude others from profiting from an invention
 - **Trade secrets** protect ideas that are typically unpatentable
 - **Trademarks** ~ recognizable sign, expression
 - **Copyrights** ~ creative works
- Many other property rights exist
 - Mostly formal (e.g., real property rights, permits, licenses)
 - Occasionally, informal

Example: Eli Lilly & Co. gross profits



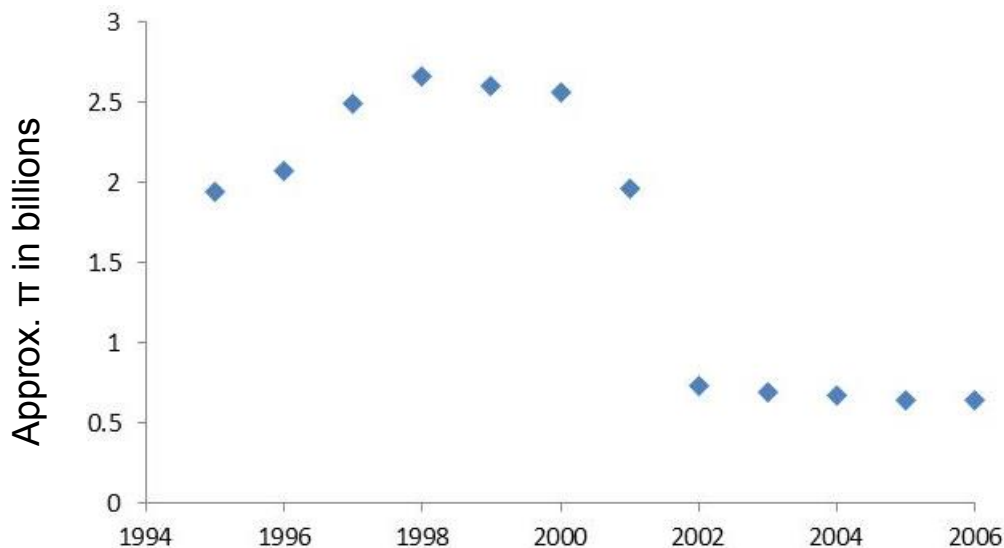
- 35,000,000 Prozac prescriptions/year as of 2000
- *What happened in that year?*

Example: Eli Lilly & Co. gross profits



- 35,000,000 Prozac prescriptions/year as of 2000
- Barr successfully challenges patent
- *How much should managers care...?*

Example: Eli Lilly & Co. gross profits



- 35,000,000 Prozac prescriptions/year as of 2000
- Barr successfully challenges patent
- LLY stock drops \$34 billion minutes after court invalidates

III. Supplier Power

The key question:

To what extent do the input prices facing a typical firm in this industry deviate from those that would prevail in perfectly competitive input markets because of the bargaining power of input suppliers?

High if

- Input is a critical component of production
- Supplier industry more concentrated than the industry it sells to.
- Firms in the analyzed industry purchase relatively small volumes relative to other customers of the supplier.
- Suppliers can credibly threaten to forward integrate.
- Switching costs are high

IV. Buyer Power

The key question:

To what extent do purchase prices in this market differ from those that would prevail in a market with a large number of fragmented buyers in which buyers act as price takers.

High if:

- input is not a critical component of production
- buyers are large and few in numbers
- input is relatively standardized
- switching costs for buyers are low
- lots of substitutes
- buyers can credibly threaten to backward integrate

Industry Definition?

Identifying substitutes depends on how “finely” you draw the boundaries of the analyzed market

- Is Amtrak (railroads) a competitor or substitute to Airlines

Key criterion: substitutability

- In Demand
 - Are key consumers easily able to substitute product in industry A for B?
- In Supply
 - Are key producers able to produce either of A and B?

Different definition of relevant industry depending on context:

- At some level it does not matter: you either have a substitute or a rival; both ways need to consider it – e.g. in CCS plastics and glass

From industry to firm

How do we identify an individual firm strategy?
use SSP (textbook) to identify

Strategy Identification

A coherent business strategy has a clear statement of:

- Objectives: *where* are we going?
- Scope: in *which* products, markets, and activities will we participate?
- Competitive advantage: *how* will we:
 - Make our products more attractive than real or potential competitors?
 - Be more efficient than real or potential competitors?
- An underlying logic: *why* will we succeed in gaining the CA and retaining it?

Logic

The logic of a strategy is evaluated by querying its

(1) Internal consistency: Does the firm's various organizational practices cohere with one another and with the purported competitive advantages?

Refer to SSP textbook.

Internal; why do you think you can do this cheaper / better than competitor?

Logic

The logic of a strategy is evaluated by querying its:

(1) Internal consistency: Does the firm's various organizational practices cohere with one another and with the purported competitive advantages?

Refer to SSP textbook.

External fit. studied about population demand, barriers, legality etc.. How do our strategy fit?

(2) External consistency:

Do they really achieve competitive advantage?

How successfully do the purported competitive advantages and supporting organizational practices address key environmental challenges?

To what extent will the competitive advantage survive competition?

So what? Why does this matter?

Forecasting

- Changes in industry structure can help predict changes in profitability

Take decisions within context-fit strategy to industry

- Success factors in industry follow from industry analysis
 - What determines value? What do consumers want?
 - Competition: what drives it, along what dimensions
 - Value chain: which are critical players? How do we avoid their capturing all value?

Improve context

- If industry is unattractive, maybe we can do something about it

Porter framework 5 forces
(since the 80s)

Rivalry
Supplier power
Substitutes
Buyer power
Entry (new)

Value Creation.

Value capture; rivalry, SP, BP, New entry

- Conceptual first (not just aiming to capture. can collaborate to create)
- Complements (something that Porter 5 dont have)
- Start by demand vs cost. understand drivers of value creation, and PIE.

Value capture of airlines.

Warren Buffet analysed that if you sum up all the profits of world airlines, the total profits is zero. They create tremendous value, but cant capture.

General discussion; Industry analysis - Airport Segment.

Analysis sequence

Value creation

- shorts vs suits. different elasticity. but overall, cyclical sector.

Car Rental Business at Airport;

- low fixed costs

Rivalry

- undifferentiated.
- only about 6-7 companies in each airport, but still highly competitive.

BTE

SP

- airport space rental. Monopoly supplier for input. Power is high.
- car supplier.

BP

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